

- 8 (a) minimum frequency for electron to be emitted (from surface)
of electromagnetic radiation / light / photons
- M1
A1 [2]
- (b) $E = hc / \lambda$ or $E = hf$ and $c = f\lambda$
- C1
- either threshold wavelength = $(6.63 \times 10^{-34} \times 3.0 \times 10^8) / (5.8 \times 10^{-19})$
= 340 nm
- or energy of 340 nm photon = 4.4×10^{-19} J
- or threshold frequency = 8.7×10^{14} Hz
- or 450 nm $\rightarrow 6.7 \times 10^{14}$ Hz
- A1
- appropriate comment comparing wavelengths / energies / frequencies
- B1
- so no effect on photo-electric current
- B1 [4]

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Section B

9 (c) (i) edges can be (clearly) distinguished

B1 [1]